

$$X(n) = e^{jn\omega} \quad -\infty < n < \infty \quad \omega \text{ is a Constant}$$

$$\begin{aligned} Y(n) &= h(n) * X(n) \\ &= \sum_{k=-\infty}^{\infty} h(k) X(n-k) \Rightarrow \sum_{k=-\infty}^{\infty} h(k) e^{j\omega(n-k)} \Rightarrow e^{jn\omega} \sum_{k=-\infty}^{\infty} h(k) e^{-jk\omega} \\ &= H(e^{j\omega}) e^{jn\omega} \quad \text{eigen Value} \quad e^{jn\omega} \cdot H(e^{j\omega}) \end{aligned}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h(k) e^{-jk\omega}$$

imaginary & Real Parts: $H(e^{j\omega}) = H_R(e^{j\omega}) + j H_I(e^{j\omega})$
 magnitude & Phase: $H(e^{j\omega}) = |H(e^{j\omega})| e^{j\phi_h(\omega)}$

$$|H(e^{j\omega})|^2 = H(e^{j\omega}) H^*(e^{j\omega}) = H_R^2(e^{j\omega}) + H_I^2(e^{j\omega})$$

$$\phi_h(\omega) = \tan^{-1} \frac{H_I(e^{j\omega})}{H_R(e^{j\omega})}$$

Example :- $X(n) = \sum_{k=1}^N \alpha_k e^{jn\omega_k}$

$$Y(n) = \sum_{k=1}^N \alpha_k \underbrace{H(e^{j\omega_k})}_{\text{Freq. response}} e^{jn\omega_k}$$

$$X(n) = \cos(n\omega_0) \quad X(n) = \frac{1}{2} e^{jn\omega_0} + \frac{1}{2} e^{-jn\omega_0}$$

$$Y(n) = \frac{1}{2} H(e^{j\omega_0}) e^{jn\omega_0} + \frac{1}{2} H(e^{-j\omega_0}) e^{-jn\omega_0}$$

$$Y(n) = \frac{1}{2} H(e^{j\omega_0}) e^{jn\omega_0} + \frac{1}{2} H^*(e^{j\omega_0}) e^{-jn\omega_0} \quad \text{Note } H(e^{-j\omega}) = H^*(e^{j\omega})$$

$$Y(n) = \text{Re} \{ H(e^{j\omega_0}) e^{jn\omega_0} \} = |H(e^{j\omega_0})| \cos(n\omega_0 + \phi_h(\omega_0))$$

Periodicity

$$\omega_0 \Rightarrow \omega_0 + 2\pi$$

$$X(n) = e^{jn\omega_0} = e^{jn(\omega_0 + 2\pi)}$$

$$H(e^{jn\omega_0}) = H(e^{j(\omega_0 + 2\pi)})$$

Symmetry

$$H(e^{-j\omega}) = H^*(e^{j\omega}) \text{ if } h(n) \text{ is real-valued}$$

, the freq. response is a Conjugate Symmetric function of freq.

Conjugate Symmetry of $H(e^{j\omega})$ implies that the real part is an even function of Frequency:

$$H_R(e^{j\omega}) = H_R(e^{-j\omega})$$

Imaginary part is Odd $H_I(e^{j\omega}) = -H_I(e^{-j\omega})$

Magnitude is even $|H(e^{j\omega})| = |H(e^{-j\omega})|$

Phase & Group delay are Odd $\phi_h(\omega) = -\phi_h(-\omega)$

$$\tau_h(\omega) = -\tau_h(-\omega)$$

EX:

$$h(n) = \alpha^n u(n)$$

α is a real number with $|\alpha| < 1$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) e^{-jn\omega} = \sum_{n=0}^{\infty} \alpha^n e^{-jn\omega} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1 - \alpha} \quad |\alpha| < 1$$

$$|H(e^{j\omega})|^2 = H(e^{j\omega}) H^*(e^{j\omega}) =$$

$$\frac{1}{1 - \alpha e^{-j\omega}} \cdot \frac{1}{1 - \alpha e^{j\omega}} = \frac{1}{1 + \alpha^2 - 2\alpha \cos \omega}$$

$$\text{Phase : } \phi_h(\omega) = \tan^{-1} \frac{H_I(e^{j\omega})}{H_R(e^{j\omega})} = \tan^{-1} \frac{-\alpha \sin \omega}{1 - \alpha \cos \omega}$$

Inverting the Freq. Response

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) e^{-jn\omega}$$

$$h(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{jn\omega} d\omega$$

EX: $H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \omega_0 \\ 0 & \omega_0 < |\omega| \leq \pi \end{cases}$

(this system is referred to as an **ideal Low-pass filter**), the unit sample response is:

$$h(n) = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} e^{jn\omega} d\omega = \frac{1}{2j\pi n} [e^{jn\omega_0} - e^{-jn\omega_0}] = \frac{\sin n\omega_0}{\pi n}$$

this system is **non Causal, Unstable & Unrealizable**.

« Filters » Lec 6 & 7

Linear Phase

$$H(e^{j\omega}) = A(e^{j\omega}) e^{-j\alpha\omega}$$

Phase of $H(e^{j\omega})$

$$\phi_h(\omega) = \begin{cases} -\alpha\omega & \text{When } A(e^{j\omega}) \geq 0 \\ -\alpha\omega + \pi & \text{When } A(e^{j\omega}) < 0 \end{cases}$$

α is real number
 $A(e^{j\omega})$ real-valued

$$H(e^{j\omega}) = A(e^{j\omega}) e^{-j(\alpha\omega - \beta)} \text{ "Generalized Linear Phase"}$$

All Pass Freq. response magnitude is "Constant"

$$|H(e^{j\omega})| = C$$

α is any real number
 $|\alpha| < 1$

example:

$$H(e^{j\omega}) = \frac{e^{-j\omega} - \alpha}{1 - \alpha e^{-j\omega}}$$

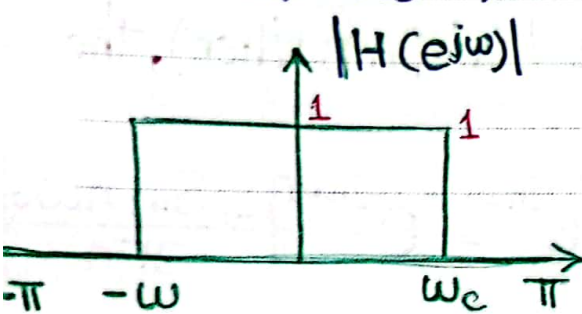
Unit Sample response of allpass filter is:

$$h(n) = -\alpha \delta(n) + (1 - \alpha^2) \alpha^{n-1} u(n-1)$$

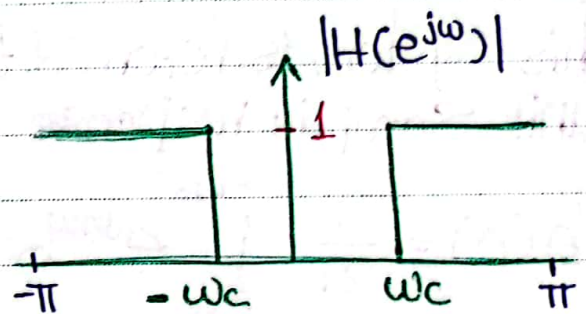
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Date :

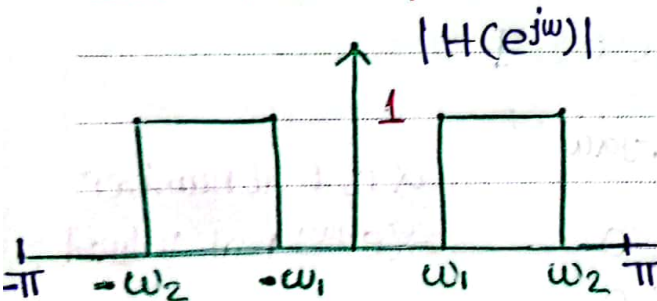
$$\begin{aligned}
 H(e^{j\omega}) &= -\alpha + \left(\frac{1-\alpha^2}{\alpha}\right) \sum_{-\infty}^{\infty} [\alpha^n u(n-1) e^{-j\omega}] = \\
 &= -\alpha + \left(\frac{1-\alpha^2}{\alpha}\right) \left[\sum_0^{\infty} (\alpha^n e^{-j\omega} - 1) \right] = \\
 &= -\cancel{\alpha} + \left(\frac{1-\alpha^2}{\alpha}\right) \cdot \frac{1}{1-\alpha e^{-j\omega}} - \frac{1}{\alpha} + \cancel{\alpha} = \frac{1-\alpha^2 - 1 + \alpha e^{-j\omega}}{\alpha(1-\alpha e^{-j\omega})} \\
 &= \frac{\alpha e^{-j\omega} - \alpha^2}{\alpha(1-\alpha e^{-j\omega})} = \frac{e^{-j\omega} - \alpha}{1-\alpha e^{-j\omega}}
 \end{aligned}$$



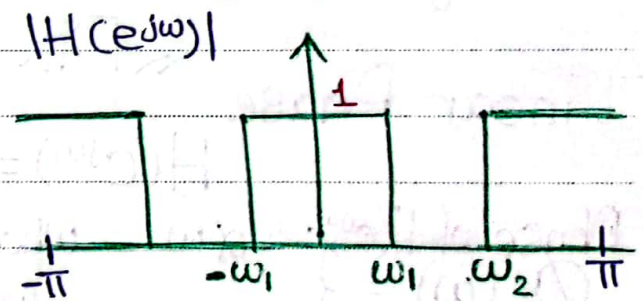
"Ideal low-pass filter"



"Ideal high-pass filter."



"Ideal bandpass filter"

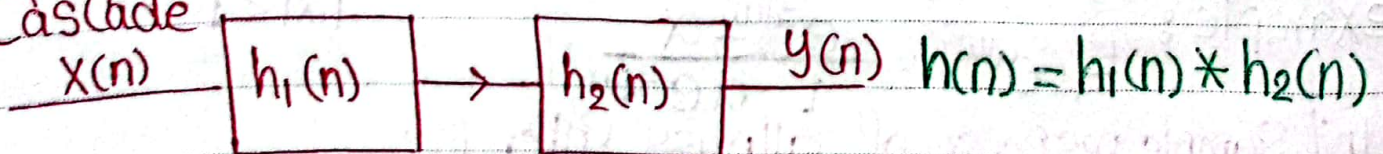


"Ideal bandstop filter"

Interconnection of Systems

2 Common types of Connections are Series (Cascade) & Parallel.

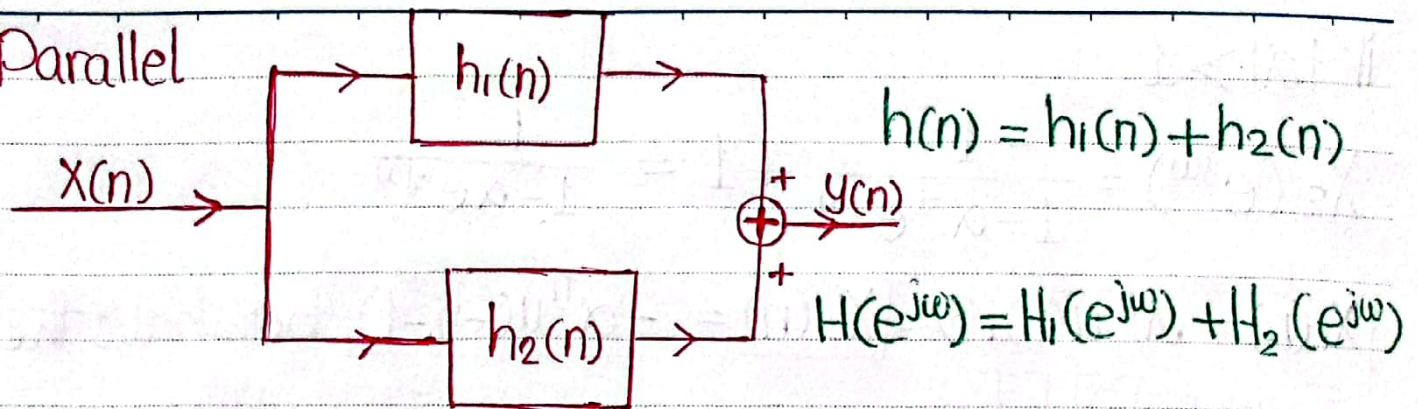
Cascade



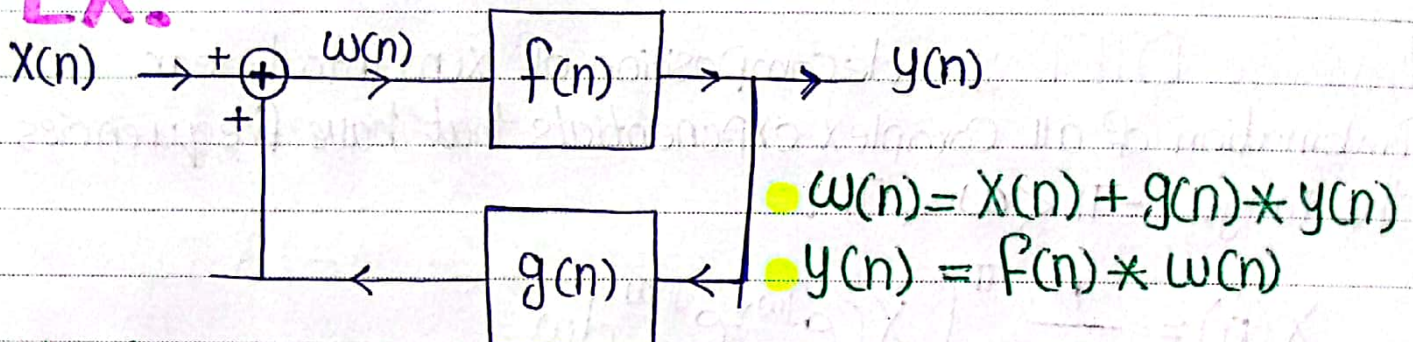
$$H(e^{j\omega}) = H_1(e^{j\omega}) H_2(e^{j\omega})$$

$$\phi_h(\omega) = \phi_{h_1}(\omega) + \phi_{h_2}(\omega)$$

Parallel



EX:



$$H(e^{j\omega}) = \frac{F(e^{j\omega})}{1 - F(e^{j\omega})G(e^{j\omega})}$$

DTFT (Discrete time Fourier transform)

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} X(n) e^{-jn\omega}$$

to exist :- $\sum_{n=-\infty}^{\infty} |X(n)| = S < \infty$

EX: $X_1(n) = \alpha^n u(n)$ $|\alpha| < 1$

$$X_1(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-jn\omega} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n$$

$$X_1(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$X_2(n) = -\alpha^n u(-n-1)$$

$$|\alpha| > 1 \Rightarrow$$

مقلی

$$X_2(e^{j\omega}) = \sum_{n=-\infty}^{\infty} X_2(n) e^{-jn\omega} = -\sum_{n=-\infty}^{\infty} \alpha^n e^{-jn\omega}$$

$$X_2(e^{j\omega}) = -\sum_{n=1}^{\infty} \alpha^{-n} e^{jn\omega} = -\sum_{n=0}^{\infty} (\alpha^{-1} e^{j\omega})^n + 1$$

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if $|\alpha| > 1$

$$X_2(e^{j\omega}) = \frac{-1}{1 - \alpha^{-1}e^{j\omega}} + 1 = \frac{1}{1 - \alpha e^{-j\omega}}$$

$X_1(n) = \alpha^n u(n)$ & $X_2(n) = -\alpha^n u(-n-1)$ both have the Same DTFT

« Inverse DTFT » decomposition of $x(n)$ into linear combination of all complex exponentials that have frequencies in the range $-\pi < \omega \leq \pi$.

$$X(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Sequence

DTFT

1. $\delta(n)$	1
2. $\delta(n-n_0)$	$e^{-jn\omega_0}$
3. 1	$2\pi \delta(\omega)$
4. $e^{jn\omega_0}$	$2\pi \delta(\omega - \omega_0)$
5. $a^n u(n)$ $ \alpha < 1$	$\frac{1}{1 - \alpha e^{-j\omega}}$
6. $-a^n u(-n-1)$ $ \alpha > 1$	$\frac{1}{1 - \alpha e^{-j\omega}}$
7. $(n+1)a^n u(n)$ $ \alpha < 1$	$\frac{1}{(1 - \alpha e^{-j\omega})^2}$
8. $\cos n\omega_0$	$\pi \delta(\omega + \omega_0) + \pi \delta(\omega - \omega_0)$

EX: $X(e^{j\omega}) = \delta(\omega - \omega_0)$

$$X(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{jn\omega} d\omega = \frac{1}{2\pi} e^{jn\omega_0} \text{ or } \frac{e^{jn\omega_0}}{2\pi}$$

EX: $X(e^{j\omega}) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$

$$X(n) = \frac{1}{2} e^{jn\omega_0} + \frac{1}{2} e^{-jn\omega_0} = \cos(n\omega_0)$$

$$\delta(\omega - \omega_0) = e^{jn\omega_0}, \quad \delta(\omega + \omega_0) = e^{-jn\omega_0} \quad \text{ملحوظة}$$

$$2\pi \delta(\omega - \omega_0) \leftarrow$$

* $2\pi \delta(\omega - \omega_0) = e^{jn\omega_0}$

* $\pi \delta(\omega - \omega_0) = \frac{1}{2} e^{jn\omega_0} / \frac{e^{jn\omega_0}}{2}$

«DTFT Properties»

Periodicity $X(e^{j\omega}) = X(e^{j(\omega + 2\pi)})$

$$X(e^{j(\omega + 2\pi)}) = \sum_{n=-\infty}^{\infty} x(n) e^{-jn(\omega + 2\pi)} = \sum_{n=-\infty}^{\infty} x(n) e^{-jn\omega} e^{-j2\pi n} =$$

$$\sum_{n=-\infty}^{\infty} x(n) e^{-jn\omega} = X(e^{j\omega})$$

Property	Sequence	DTFT
Linearity	$aX(n) + bY(n)$	$\Leftrightarrow aX(e^{j\omega}) + bY(e^{j\omega})$
Shift	$X(n - n_0)$	$\Leftrightarrow e^{-jn\omega_0} X(e^{j\omega})$
Time-reversal	$X(-n)$	$\Leftrightarrow X(e^{-j\omega})$
Modulation	$e^{jn\omega_0} X(n)$	$\Leftrightarrow X(e^{j(\omega - \omega_0)})$
Convolution	$X(n) * Y(n)$	$\Leftrightarrow X(e^{j\omega}) Y(e^{j\omega})$
Conjugation	$X^*(n)$	$\Leftrightarrow X^*(e^{-j\omega})$
Derivative	$n X(n)$	$\Leftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$
Multiplication	$X(n) Y(n)$	$\Leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\omega - \theta)}) d\theta$

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Symmetry

 $x(n)$

- Real & even
- Real & odd
- Imaginary & even
- Imaginary & odd

 $X(e^{j\omega})$

- Real & even
- Imaginary & odd
- Imaginary & even
- Real & odd

Linearity

$$aX_1(n) + bX_2(n) \xrightarrow{\text{DTFT}} aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

Shifting

$$X(n-n_0) \xrightarrow{\text{DTFT}} e^{-jn_0\omega} X(e^{j\omega})$$

Time-Reversal

$$X(-n) \xrightarrow{\text{DTFT}} X(e^{-j\omega})$$

Modulation

$$e^{jn\omega_0} X(n) \xrightarrow{\text{DTFT}} X(e^{j(\omega-\omega_0)})$$

$$X(n) \cos n\omega_0 \xrightarrow{\text{DTFT}} \frac{1}{2} X(e^{j(\omega-\omega_0)}) + \frac{1}{2} X(e^{j(\omega+\omega_0)})$$

Convolution

$$h(n) * X(n) \xrightarrow{\text{DTFT}} H(e^{j\omega}) X(e^{j\omega})$$

Multiplication (Periodic Convolution) theorem

$$X(n) Y(n) \xrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

Parseval's Theorem (Conservation of energy theorem)

$$\sum_{n=-\infty}^{\infty} |X(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

energy in time domain = energy in Frequency domain

Applications

1. LSI Systems & LCCDEs

$$y(n) = - \sum_{k=1}^p a(k) y(n-k) + \sum_{k=0}^q b(k) x(n-k)$$

$$Y(e^{j\omega}) = - \sum_{k=1}^p a(k) e^{-jk\omega} Y(e^{j\omega}) + \sum_{k=0}^q b(k) e^{-jk\omega} X(e^{j\omega})$$

or $Y(e^{j\omega}) \left[1 + \sum_{k=1}^p a(k) e^{-jk\omega} \right] = X(e^{j\omega}) \sum_{k=0}^q b(k) e^{-jk\omega}$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^q b(k) e^{-jk\omega}}{1 + \sum_{k=1}^p a(k) e^{-jk\omega}}$$

Ex: $y(n) = 1.3433y(n-1) - 0.9025y(n-2) + x(n) - 1.4142x(n-1) + x(n-2)$

$$H(e^{j\omega}) = \frac{1 - 1.4142e^{-j\omega} + e^{-2j\omega}}{1 - 1.3433e^{-j\omega} + 0.9025e^{-2j\omega}}$$

Ex:

$$H(e^{j\omega}) = \frac{1 + e^{-2j\omega}}{2 - e^{-j\omega} + 0.5e^{-2j\omega}}$$

يمكن تبسيط السؤال بالأكس!

$$H(e^{j\omega}) = \frac{0.5 + 0.5e^{-2j\omega}}{1 - 0.5e^{-j\omega} + 0.25e^{-2j\omega}}$$

$$y(n) = 0.5y(n-1) - 0.25y(n-2) + 0.5x(n) + 0.5x(n-2)$$

Performing Convolutions

"Convolution in time domain $\Rightarrow$ Multiplication in Freq. domain."

Ex:

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$$h(n) = \alpha^n u(n)$$

$$|\alpha| < 1$$

$$x(n) = \beta^n u(n)$$

$$|\beta| < 1$$

$$\alpha \neq \beta$$

$$y(n) = h(n) * x(n)$$

$$Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}} \cdot \frac{1}{1 - \beta e^{-j\omega}}$$

$$Y(e^{j\omega}) = \frac{1}{(1 - \alpha e^{-j\omega})(1 - \beta e^{-j\omega})} = \frac{A}{1 - \alpha e^{-j\omega}} + \frac{B}{1 - \beta e^{-j\omega}}$$

$$\frac{1}{(1 - \alpha e^{-j\omega})(1 - \beta e^{-j\omega})} = \frac{(A+B) - (A\beta + B\alpha)e^{-j\omega}}{(1 - \alpha e^{-j\omega})(1 - \beta e^{-j\omega})} = \frac{1}{(1 - \alpha e^{-j\omega})(1 - \beta e^{-j\omega})}$$

$$A+B = 1 \quad \text{Real} \quad A\beta + B\alpha = 0 \quad \text{Imaginary}$$

$$A = \frac{\alpha}{\alpha - \beta}$$

$$B = -\frac{\beta}{\alpha - \beta}$$

$$Y(e^{j\omega}) = \frac{\alpha / (\alpha - \beta)}{1 - \alpha e^{-j\omega}} - \frac{\beta / (\alpha - \beta)}{1 - \beta e^{-j\omega}}$$

$$Y(n) = \left[\frac{\alpha}{\alpha - \beta} \alpha^n - \frac{\beta}{\alpha - \beta} \beta^n \right] u(n)$$

$$y(n) = [A \alpha^n + B \beta^n] u(n) \quad \text{"Inverse DTFT"}$$

EX: $y(n) - 0.25y(n-1) = x(n) - x(n-2)$

$$x(n) = \delta(n)$$

Assume $\rightarrow$

$$X(e^{j\omega}) = 1$$

$$y(e^{j\omega}) - 0.25e^{-j\omega} y(e^{j\omega}) = X(e^{j\omega}) - e^{-2j\omega} X(e^{j\omega})$$

$$y(e^{j\omega}) = \frac{1 - e^{-2j\omega}}{1 - 0.25e^{-j\omega}} = \frac{1}{1 - 0.25e^{-j\omega}} - \frac{e^{-2j\omega}}{1 - 0.25e^{-j\omega}}$$

$$(0.25)^n u(n) \xleftrightarrow{\text{DTFT}}$$

$$\frac{1}{1 - 0.25e^{-j\omega}}$$

$$y(n) = (0.25)^n u(n) - (0.25)^{n-2} u(n-2) \quad \neq$$

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Inverse Systems

$$h(n) * g(n) = \delta(n)$$

$$G(e^{j\omega}) = \frac{1}{H(e^{j\omega})} = \frac{1}{1 - 2e^{-j\omega}}$$

$$H(e^{j\omega}) = 1 - 2e^{-j\omega}$$

$$g(n) = -2^{-n} u(-n-1)$$

EX: $H(e^{j\omega}) = \frac{1 - \frac{1}{4}e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$

$$G(e^{j\omega}) = \frac{1}{H(e^{j\omega})} = \frac{1 + \frac{1}{2}e^{-j\omega}}{1 - \frac{1}{4}e^{-j\omega}}$$

$$g(n) = (0.25)^n u(n) + 0.5(0.25)^{n-1} u(n-1)$$